

I - Contextualization

Chapter 1
From Speed Recommendation to Adaptive Control

II - State of the Art

Chapter 2.1
How can a learning-based controller adapt continuously to non-stationary dynamics while maintaining reliability and safety ?

Chapter 2.2
How can a learning architecture model non-linear, non-stationary dynamics from an online stream of data while avoiding catastrophic forgetting ?

III - Contributions

Chapter 4
Solving Control Tasks with CELL
(Model Predictive Control)

Chapter 3
Context Ensemble Local Learning
(CELL formalism $\rightarrow$ oCELL $\rightarrow$ kCELL)

Chapter 5
Prospective works on predictive uncertainty quantification and usage of CELL modularity
(CELL formalism $\rightarrow$ gpCELL)

IV - Conclusions and Future Research Directions

Chapter 6
Industrial Use Cases

Chapter 7
Conclusion and Future Works

Control with
Learned Models

Online Learning